

Notes: Ellison (QJE, 2005)

A Model of Add-On Pricing

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1 Big picture

- **Question.** Why do firms in competitive markets advertise low base prices but charge high prices for add-ons, and what does this practice do to competition, profits, and consumer welfare?
- **Core examples.** Hotel rooms plus phone calls, minibar items, and meals; personal computers plus memory or video-card upgrades; appliances plus extended warranties; car rentals plus insurance and prepaid gasoline; homes plus upgrades.
- **Key distinction.** The paper distinguishes ordinary second-degree price discrimination, where all prices are observable, from add-on pricing, where base prices are advertised but add-on prices are naturally hard or costly to observe before visiting the seller.
- **Main theoretical environment.** Two horizontally differentiated firms compete on a Hotelling line and offer a low-quality/base product L and a high-quality/bundled product H . Consumers differ both horizontally and vertically.
- **Consumer heterogeneity.** High types have lower marginal utility of income, value quality more in money terms, and are less price-sensitive across firms. Low types, or “cheapskates,” are more price-sensitive and less willing to buy high-priced add-ons.
- **Benchmark irrelevance result.** If high and low types are similar enough, add-on pricing is irrelevant. Firms sell the high-quality product to everyone, and hidden add-on prices simply become part of the total price.
- **Main add-on-pricing result.** When types differ more, add-on pricing softens price competition. Hidden high add-on prices make price cuts unattractive because a firm that cuts the advertised base price disproportionately attracts low-margin cheapskates.
- **Mechanism.** Add-on pricing creates an adverse-selection problem in customer acquisition. Price-cutting attracts consumers who buy only the low-priced base good and avoid the profitable add-on.
- **Profit implication.** Joint adoption of add-on pricing can raise equilibrium profits relative to standard observable price discrimination.
- **Welfare implication.** High types are hurt by high add-on prices, low types may benefit from lower base prices, and a regulation forcing add-ons to be free can sometimes make all consumers better off.
- **Adoption puzzle.** Although add-on pricing can raise joint profits, it may not be individually stable when firms can costlessly advertise add-on prices. A firm can profitably advertise a lower add-on price to steal high types.
- **Cheapskate externality.** In a version with a small number of cheapskates, their presence can raise firms’ profits under add-on pricing because firms use cheapskates to soften competition for high types.
- **Takeaway.** Hidden add-on prices can be a competition-softening device in competitive markets, not merely a way to extract rents from inattentive consumers.

2 Incremental contribution of the paper

2.1 Contribution relative to the Chicago-school benchmark

- The standard Chicago-school view says profits from add-ons should be competed away through lower base prices in competitive markets.
- Ellison shows this view is correct only in special cases, such as when everyone buys the add-on or when product/pricing assumptions eliminate selection effects.

- The paper identifies a precise reason why add-on profits may not be fully competed away: a lower base price attracts a worse mix of customers.

Interpretation: The incremental contribution is not simply “add-ons are expensive.” The paper explains why expensive add-ons can raise total equilibrium profits even when firms compete intensely for customers.

2.2 Contribution relative to loss-leader models

- Lal and Matutes-style loss-leader models show that hidden or unadvertised add-on prices may have no effect when all consumers buy the same bundle.
- Ellison nests that logic in Proposition 1 but then shows that it breaks down when add-on demand differs systematically across consumers.
- The paper therefore transforms an irrelevance benchmark into a theory of when add-on pricing matters.

Interpretation: The contribution is a conditional theory: add-ons are irrelevant when they are effectively bundled for all consumers, but matter when they separate price-sensitive from profitable consumers.

2.3 Contribution relative to competitive price discrimination

- Competitive second-degree price discrimination models can be complex because horizontal and vertical differentiation interact.
- Ellison provides a tractable Hotelling model in which discriminatory and nondiscriminatory equilibria can coexist.
- The multiplicity shows that price discrimination can be self-reinforcing: discriminating is more attractive when rivals discriminate.

Interpretation: The paper contributes a simple model of competitive second-degree price discrimination with clear equilibrium comparisons and an intuitive multiplicity result.

2.4 Contribution relative to behavioral add-on theories

- Behavioral explanations emphasize unsophisticated consumers, shrouded attributes, or mistakes.
- Ellison’s central competition-softening result does not require irrational expectations in the equilibrium analysis. Consumers rationally anticipate hidden prices.
- Behavioral ingredients become important mainly for explaining why firms do not advertise add-on prices when advertising is possible.

Interpretation: The paper’s main result is robustly rational: even sophisticated consumers can be in an equilibrium where hidden add-ons soften competition through selection.

2.5 Contribution to industrial organization more broadly

- The paper highlights that the profitability of a low base price depends on customer composition, not just total demand elasticity.
- It shows how multiproduct pricing can make marginal customers unprofitable even when the market is competitive.
- It provides a theory for why competitive industries with apparently small differentiation may sustain markups through add-on practices.

Interpretation: The key IO lesson is that firms may strategically create add-on pricing systems that make aggressive base-price competition less attractive.

3 Model setup

3.1 Products and firms

There are two firms indexed by $i \in \{1, 2\}$ located at opposite ends of a Hotelling line. Each firm sells two vertically differentiated goods:

- L : the base or low-quality product;
- H : the high-quality product, interpreted as the base product plus an add-on.

Both goods have the same constant marginal cost c . This assumption captures a damaged-goods logic: good L is not cheaper to produce but is less valuable to consumers.

Interpretation: Equal cost is important. If L were cheaper to produce, low-type purchases of L would not be as harmful to profits. The model instead makes L a low-margin way to serve cheapskates.

3.2 Consumers

There are two unit masses of consumers. They differ in marginal utility of income:

$$\alpha_h < \alpha_l. \quad (1)$$

High types h have lower marginal utility of income and thus higher willingness to pay for quality. Low types l are more price-sensitive and are called cheapskates.

Within each type group, consumers are indexed by a horizontal taste parameter

$$\theta \sim U[0, 1]. \quad (2)$$

A type (α, θ) consumer's utility from buying exactly one product is

$$u(q_{1L}, q_{1H}, q_{2L}, q_{2H}; \alpha, \theta) = \begin{cases} v - \theta - \alpha p_{1H}, & \text{if } q_{1H} = 1, \\ v - (1 - \theta) - \alpha p_{2H}, & \text{if } q_{2H} = 1, \\ v - w - \theta - \alpha p_{1L}, & \text{if } q_{1L} = 1, \\ v - w - (1 - \theta) - \alpha p_{2L}, & \text{if } q_{2L} = 1. \end{cases} \quad (3)$$

Here w is the utility loss from consuming L rather than H .

Interpretation: The same difference in utility, w , translates into a larger money value for high types because they have lower α . Thus high types are more willing to pay for add-ons and are less price-sensitive across firms.

3.3 Two pricing games

The paper compares two games.

1. **Standard pricing game.** Firms simultaneously post both prices p_{iL} and p_{iH} . Consumers observe all prices before choosing a firm and a product.
2. **Add-on pricing game.** Firms post only p_{iL} initially. After consumers visit a firm, firms reveal the upgrade/add-on price $p_{iH} - p_{iL}$. Consumers form rational expectations about hidden prices but may not observe deviations unless they visit.

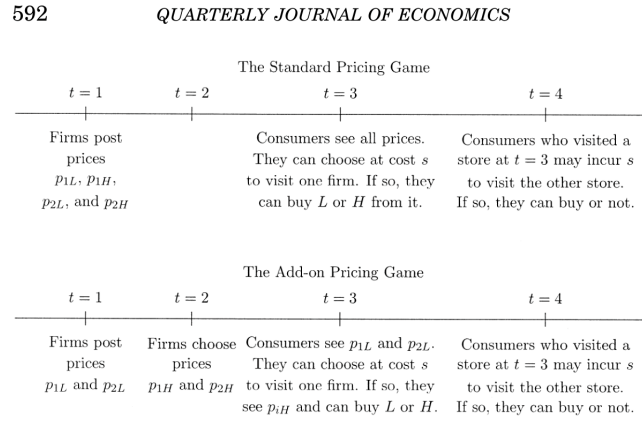
Interpretation: The add-on game captures hotels, rental cars, warranties, credit cards, and other markets in which base prices are advertised but add-on prices are difficult to compare before shopping.

3.4 Timeline

Read:

- Figure I contrasts the standard pricing game and the add-on pricing game.
- In the standard game, all prices are posted before consumers search.
- In the add-on game, firms post low-product prices first and choose the hidden high-product prices later.
- Consumers visit a firm, see the nonposted high-price information, and may pay a search cost to switch firms.

Economic message: The timing is what gives hidden add-on prices their strategic force. Add-on deviations are not immediately visible to all consumers.



4 Benchmark: when add-ons are irrelevant

4.1 Logic of the Lal–Matutes benchmark

When high and low types are similar enough, both groups buy the high-quality product. In that case, hidden add-on pricing is economically similar to adding an unadvertised fee that everyone expects and everyone pays.

- Firms' advertised base prices adjust downward to offset expected add-on payments.
- Consumers buy the same total product bundle.
- Firms earn the same profits as when all prices are observable.

Interpretation: Hidden add-ons are irrelevant when they do not change consumer sorting across products.

4.2 Proposition 1

Read:

- Proposition 1 assumes $\alpha_l/\alpha_h \leq 1.6$.
- In any symmetric pure-strategy equilibrium of the standard pricing game, all consumers buy H from the closest firm at price $c + 1/\bar{\alpha}$.

- In any symmetric pure-strategy equilibrium of the add-on pricing game, the same total outcome occurs.

Economic message: This proposition formalizes the Chicago-style irrelevance result: when everyone buys the add-on, add-on profits are fully competed away.

ment in the text leaving the details to the Appendix.

PROPOSITION 1. Suppose that $\alpha_l/\alpha_h \leq 1.6$. Write $\bar{\alpha}$ for $(\alpha_l + \alpha_h)/2$.

Then, for v sufficiently large, the standard- and add-on-pricing games have pure strategy equilibria and

- (a) In any symmetric pure-strategy sequential equilibrium of the standard pricing game, all consumers buy the high-quality good from the closest firm at a price of $c + 1/\bar{\alpha}$.
- (b) In any symmetric pure-strategy sequential equilibrium of the add-on pricing game, all consumers buy the high-quality good from the closest firm at a price of $c + 1/\bar{\alpha}$.

5 Standard competitive price discrimination

5.1 Why price discrimination emerges

When types differ more, the low and high types are sufficiently distinct that firms may prefer to sell L to low types and H to high types.

- Low types are price-sensitive and willing to accept the lower-quality good.
- High types value quality more and are willing to buy the high-quality good.
- Firms can compete almost separately for each segment.

Interpretation: Competitive second-degree price discrimination becomes feasible because the incentive constraints are not binding for the relevant parameter range.

5.2 Proposition 2

Read:

- Proposition 2 assumes $\alpha_l/\alpha_h \in [3.2, 10]$ and a quality gap w in an intermediate interval.
- The standard pricing game has a discriminatory equilibrium: low types buy L at price $c + 1/\alpha_l$ and high types buy H at price $c + 1/\alpha_h$.
- For sufficiently large α_l/α_h , the standard pricing game may also have a nondiscriminatory equilibrium in which everyone buys H at price $c + 1/\bar{\alpha}$.

Economic message: Price discrimination is self-reinforcing. A firm's gain from discriminating is larger when its rival also discriminates.

are farther apart, and it is optimal to discriminate.

PROPOSITION 2. Suppose that $\alpha_l/\alpha_h \in [3.2, 10]$. Let $w = \alpha_h(1/\alpha_h - 1/\alpha_l)$. Let $\bar{w} = 4(\bar{\alpha}/\sqrt{\alpha_l\alpha_h} - 1)$. Then $\bar{w} > w$ and for $w \in (w, \bar{w})$,

- (a) The standard pricing game has a “discriminatory” sequential equilibrium in which the low types buy good L from the closest firm at a price of $c + 1/\alpha_l$ and the high types buy good H from the closest firm at a price of $c + 1/\alpha_h$.
- (b) If $\alpha_l/\alpha_h > 6.4$ (and for some other parameter values), the standard pricing game also has a sequential equilibrium in

6 Add-on pricing and softened competition

6.1 Why hidden add-on prices change competition

In the add-on pricing game, firms compete on the advertised low-product price p_L , but the profitable upgrade to H is unadvertised. Hidden add-on prices are set high because once a consumer has visited a firm, search costs are sunk.

- High types expect to buy the add-on and generate high margins.
- Low types buy only the low-priced base good and generate low margins.
- A cut in the advertised base price disproportionately attracts low types.
- This adverse-selection effect weakens the incentive to cut prices.

Interpretation: Add-on pricing transforms price competition into a problem of attracting the right mix of customers.

6.2 Proposition 3

Read:

- Proposition 3 studies the add-on pricing game in the same parameter range as Proposition 2.
- There is a sequential equilibrium in which firms set a low advertised base price.
- Low types buy L from the closest firm.
- High types pay a high upgrade price to buy H .
- Equilibrium profits can exceed those in the standard pricing game.

Economic message: The core result is that hidden add-ons can raise profits in competitive markets by reducing firms’ desire to undercut base prices.

price of w/α_l . Part (a) describes the equilibrium that seems most reasonable. Part (b) notes another possibility that one could imagine might also arise in some industries—an expectations trap in which consumer beliefs that add-ons will be sold at low prices make it impossible for firms to charge high prices.

PROPOSITION 3. Suppose that $\alpha_l/\alpha_h \in [3.2, 10]$ and $w \in (w, \bar{w})$. Then,

- (a) The add-on pricing game has a sequential equilibrium in which the firms set $p_{iL} = c + 1/\bar{\alpha} - w/2\bar{\alpha}$, low types buy

6.3 Adverse selection in base-price competition

Let a firm's expected margin from a consumer be higher for high types than for low types. If lowering p_L draws disproportionately many low types, the marginal customer is less profitable than the average customer. Therefore a base-price cut is less attractive.

Interpretation: This is analogous to adverse selection in insurance. Better coverage attracts worse risks; lower base prices attract cheapskates.

6.4 Welfare consequences

Compared with standard observable price discrimination:

- high types are hurt because they pay more for add-ons;
- low types may benefit because advertised base prices are lower;
- firms earn higher profits;
- regulations forcing free add-ons can sometimes make all consumers better off.

Interpretation: A restriction that seems to remove consumer choice can be welfare-improving because it prevents firms from using hidden add-ons to soften competition.

7 Endogenous advertising and adoption of add-on pricing

7.1 The adoption puzzle

If firms can costlessly advertise add-on prices, then the high-profit add-on equilibrium may not be individually stable. A firm can advertise a slightly lower add-on price and steal profitable high types from its rival.

Interpretation: Joint adoption of add-on pricing can raise profits, but individual firms may have incentives to reveal add-on prices if doing so steals high-margin customers.

7.2 Figure II and Proposition 4

Read:

- Figure II describes an endogenous advertising game in which firms first choose whether to post both prices, only one price, or neither.
- Proposition 4 states that the endogenous advertising game does not have an equilibrium in which firms simply play as in the add-on pricing equilibrium.

Economic message: The result explains why the prevalence of add-on pricing requires additional frictions: advertising add-on prices must be difficult, ineffective, or strategically costly.

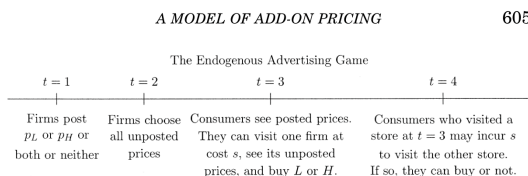


Figure II: endogenous advertising game

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rival is playing as in the add-on pricing equilibrium, then the firm will have an incentive to advertise both prices and move them closer together.

PROPOSITION 4. Suppose that $\alpha_l/\alpha_h \in [3.2, 10]$ and $w \in (w, \bar{w})$. Then, the endogenous advertising game does not have an equilibrium in which firms play as in the add-on pricing

Proposition 4: hidden add-ons are not individually stable with costless advertising

7.3 Possible explanations for observed add-on pricing

The paper discusses several explanations for why firms nevertheless use add-on pricing:

- add-on prices may be hard to communicate or compare;
- quality-adjusted add-on prices may be difficult to advertise compactly;
- consumers may be behavioral or inattentive;
- advertising add-on prices may inform irrational consumers and reduce their willingness to buy;
- norms or industry practices may coordinate firms on hidden add-on pricing.

Interpretation: The main model shows why add-on pricing is profitable when adopted. Section V asks why adoption itself persists when advertising choices are endogenous.

8 The cheapskate externality

8.1 Motivation

Section VI studies a variant in which cheapskates are a small fraction of the population. This setting clarifies how add-on pricing can make firms benefit from the presence of low-margin customers.

- In a one-good market, cheapskates intensify competition and reduce profits.
- In the add-on-pricing market, cheapskates can soften competition by making price cuts less attractive.
- When the quality gap is large enough, the ordinary intuition is reversed.

Interpretation: Cheapskates impose an externality on high types. Their presence can discipline firms away from aggressive base-price competition.

8.2 Proposition 5

Read:

- Proposition 5 contrasts a one-good cheapskate model with the cheapskate version of the add-on pricing model.
- In the one-good model, adding cheapskates lowers prices and profits.
- In the add-on model, for sufficiently large quality gap w , prices and profits can increase with the fraction of cheapskates.

Economic message: The proposition formalizes the cheapskate externality: price-sensitive consumers can raise firms' profits by making price cuts attract unprofitable customers.

L at $c + 1/\alpha_h - w/\alpha_h$, they would be selling L for less than $c + 1/\alpha_l$ and hence would prefer to serve fewer cheapskates at a higher margin.

PROPOSITION 5. Suppose that $\alpha_l/\alpha_h > 2$. Define $\bar{\alpha}_\epsilon \equiv (\alpha_h + \epsilon\alpha_l)/(1 + \epsilon)$.

(a) In the one-good version of the cheapskate model obtained

15. As in Propositions 2 and 3 the requirement is that the upgrade price w/α_h be larger than what the difference between p_H and p_L would be if the firms competed separately for the low and high types.

Proposition 5, part 1

by setting $w = 0$, for sufficiently small ϵ the unique symmetric equilibrium has $p^* = c + 1/\bar{\alpha}_\epsilon$, and prices and profits are decreasing in ϵ .

(b) If $w > w$, then for sufficiently small ϵ the unique symmetric equilibrium of the cheapskate version of the add-on pricing model has

$\frac{1}{1 + \epsilon} (w + \frac{1}{1 + \epsilon}) \leq \epsilon\alpha_l$
Proposition 5, part 2

9 Identification and empirical interpretation

9.1 The paper is theoretical

- The paper does not estimate causal effects using data.
- Its identification is theoretical: it isolates mechanisms within a model and compares equilibrium outcomes across games.
- The main comparison is between the standard pricing game and the add-on pricing game holding preferences and costs fixed.

Interpretation: The paper's goal is not to measure add-on pricing in one industry; it is to show a mechanism that can operate across many competitive markets.

9.2 What the model explains well

- high advertised markups on add-ons in competitive industries;
- low base prices combined with expensive extras;
- why firms may not compete away add-on profits;
- why restricting add-on charges can sometimes benefit all consumers;
- why price-sensitive consumers can make competition less aggressive.

Interpretation: The model is most relevant when consumers who buy add-ons are also less price-sensitive in choosing among firms.

9.3 Limits of the model

- It assumes rational expectations about hidden add-on prices.
- It relies on a positive correlation between willingness to buy add-ons and brand loyalty or lower price sensitivity.
- It uses a two-firm Hotelling framework and a simple two-type consumer population.
- It does not fully model the behavioral or institutional reasons why add-on prices remain hard to observe.
- The endogenous advertising section shows that additional frictions are needed to rationalize persistent hidden add-on pricing.

Interpretation: The paper's core mechanism is powerful but not universal. It applies best to markets where high-margin add-on buyers are also infra-marginal consumers.

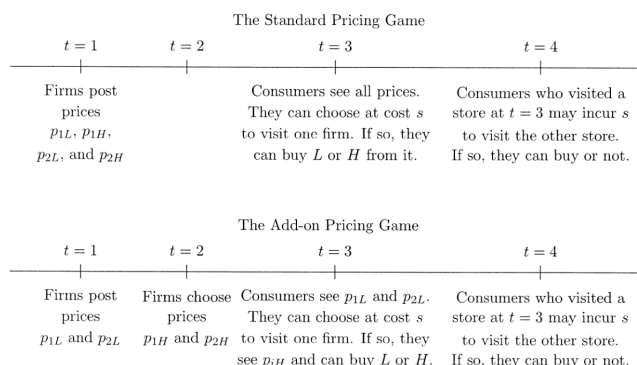
10 Figures and proposition panels

10.1 Figure I: timing of standard and add-on pricing

Read:

- The standard pricing game posts all prices at the start.
- The add-on pricing game posts only base-good prices first and reveals high-good prices only after consumers visit a firm.
- The later revelation of add-on prices creates the search friction that supports the model.

Economic message: Timing is the difference between ordinary price discrimination and add-on pricing. Hidden timing generates the competition-softening channel.



10.2 Propositions 1–3: benchmark, standard discrimination, and add-on pricing

Read:

- Proposition 1: when types are similar, all consumers buy H and add-on pricing has no effect.
- Proposition 2: when types differ more, standard observable pricing can have discriminatory and nondiscriminatory equilibria.
- Proposition 3: with hidden add-on prices, firms can earn higher profits because price cuts attract too many low-margin cheapskates.

Economic message: These propositions provide the paper’s main sequence: irrelevance, competitive price discrimination, and competition-softening add-on pricing.

ment in the text leaving the details to the Appendix.

PROPOSITION 1. Suppose that $\alpha_l/\alpha_h \leq 1.6$. Write $\bar{\alpha}$ for $(\alpha_l + \alpha_h)/2$.

Then, for v sufficiently large, the standard- and add-on-pricing games have pure strategy equilibria and

(a) In any symmetric pure-strategy sequential equilibrium of the standard pricing game, all consumers buy the high-quality good from the closest firm at a price of $c + 1/\bar{\alpha}$.

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are farther apart, and it is optimal to discriminate.

PROPOSITION 2. Suppose that $\alpha_l/\alpha_h \in [3.2, 10]$. Let $w = \alpha_h(1/\alpha_h - 1/\alpha_l)$. Let $\bar{w} = 4(\bar{\alpha}/\sqrt{\alpha_l\alpha_h} - 1)$. Then $\bar{w} > w$ and for $w \in (w, \bar{w})$,

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(b) If $\alpha_l/\alpha_h > 6.4$ (and for some other parameter values), the standard pricing game also has a sequential equilibrium in

Proposition 2: standard pricing game

price of $c + 1/\alpha_l$. Part (a) describes the equilibrium that seems most reasonable. Part (b) notes another possibility that one could imagine might also arise in some industries—an expectations trap in which consumer beliefs that add-ons will be sold at low prices make it impossible for firms to charge high prices.

PROPOSITION 3. Suppose that $\alpha_l/\alpha_h \in [3.2, 10]$ and $w \in (w, \bar{w})$.

Then,

(a) The add-on pricing game has a sequential equilibrium in which the firms set $p_{iL} = c + 1/\bar{\alpha} - w/2\bar{\alpha}$, low types buy

Proposition 3: add-on pricing game

10.3 Figure II and Proposition 4: endogenous advertising

Read:

- Figure II adds a first-stage advertising choice.

- Proposition 4 shows that the add-on equilibrium is not stable when firms can costlessly advertise add-on prices.

Economic message: The theory of profitability is not yet a full theory of adoption. Additional communication or behavioral frictions are needed.

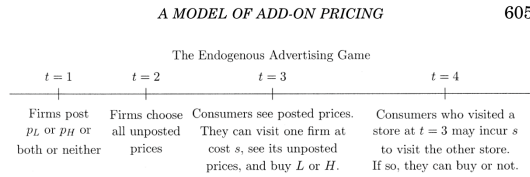


Figure II: endogenous advertising

rival is playing as in the add-on pricing equilibrium, then the firm will have an incentive to advertise both prices and move them closer together.

PROPOSITION 4. Suppose that $\alpha_l/\alpha_h \in [3.2, 10]$ and $w \in (\underline{w}, \bar{w})$. Then, the endogenous advertising game does not have an equilibrium in which firms play as in the add-on pricing

Proposition 4: advertising deviation

10.4 Proposition 5: cheapskate externality

Read:

- Proposition 5 compares a one-good market with the add-on pricing market when cheapskates are a small group.
- In the one-good market, cheapskates reduce profits.
- In the add-on market, cheapskates can increase prices and profits when the quality gap is large.

Economic message: Price-sensitive consumers can make competition less aggressive by making low base-price deviations unattractive.

L at $c + 1/\alpha_h - w/\alpha_h$, they would be selling L for less than $c + 1/\alpha_l$ and hence would prefer to serve fewer cheapskates at a higher margin.

PROPOSITION 5. Suppose that $\alpha_l/\alpha_h > 2$. Define $\bar{\alpha}_\epsilon \equiv (\alpha_h + \epsilon\alpha_l)/(1 + \epsilon)$.

(a) In the one-good version of the cheapskate model obtained

15. As in Propositions 2 and 3 the requirement is that the upgrade price w/α_h be larger than what the difference between p_H and p_L would be if the firms competed separately for the low and high types.

Proposition 5, part 1

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by setting $w = 0$, for sufficiently small ϵ the unique symmetric equilibrium has $p^* = c + 1/\bar{\alpha}_\epsilon$, and prices and profits are decreasing in ϵ .

(b) If $w > \underline{w}$, then for sufficiently small ϵ the unique symmetric equilibrium of the cheapskate version of the add-on pricing model has

$\frac{1}{\alpha_l} (w + \frac{1}{\alpha_h}) \backslash \backslash \epsilon \alpha_l$
Proposition 5, part 2

11 Short summary

- The paper studies why firms use low advertised base prices and high unadvertised add-on prices.
- The model has two firms, two products, two consumer types, and Hotelling horizontal differentiation.
- High types value quality more and are less price-sensitive.
- Low types or cheapskates are more price-sensitive and buy add-ons less often.
- If both types are similar, add-on pricing is irrelevant because everyone buys the high-quality bundle.
- If types differ, standard pricing can generate second-degree price discrimination.
- Hidden add-on pricing raises profits by making base-price cuts attract low-margin cheapskates.
- This is an adverse-selection channel in customer acquisition.
- High types lose from add-on pricing, low types may gain, and firms gain.
- Free-add-on regulation can sometimes improve welfare for all consumers.

- Add-on pricing is not individually stable with costless add-on-price advertising, so persistence requires communication costs or behavioral frictions.
- Cheapskates can increase firms' profits in add-on markets by softening competition.

12 Conclusion

12.1 Core conclusion

Ellison shows that hidden add-on pricing can raise profits in competitive markets by softening base-price competition. The mechanism is adverse selection: cutting the advertised base price disproportionately attracts price-sensitive consumers who do not buy profitable add-ons.

12.2 Economic intuition

A firm normally wants to cut its price to steal customers. But in an add-on market, stealing customers is valuable only if the new customers buy high-margin add-ons. If a price cut attracts mostly cheapskates, it lowers the average profitability of the customer pool. Firms therefore cut less aggressively, and equilibrium profits rise.

12.3 Takeaway

The paper's main lesson is that add-ons are not just hidden fees. They can be part of a competitive strategy that changes the composition of customers attracted by base-price competition. This helps explain why competitive markets may display low advertised prices, high add-on prices, and persistent markups.